

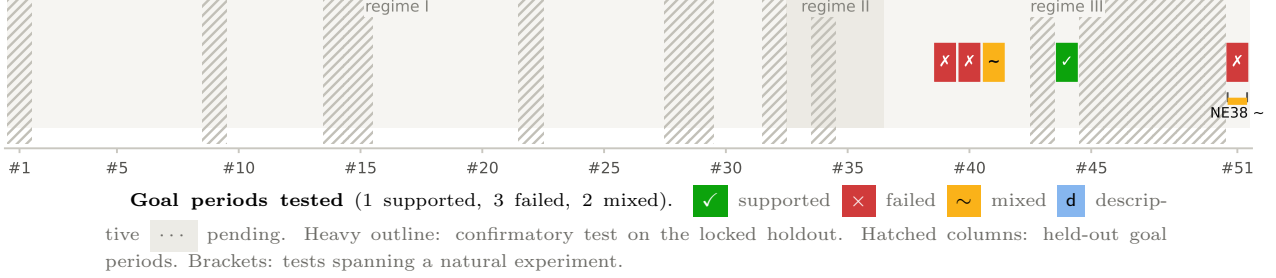
# H75: A thermodynamic speed limit on re-allocation

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 40% Faithfulness 2/5 Usefulness 2.5/5

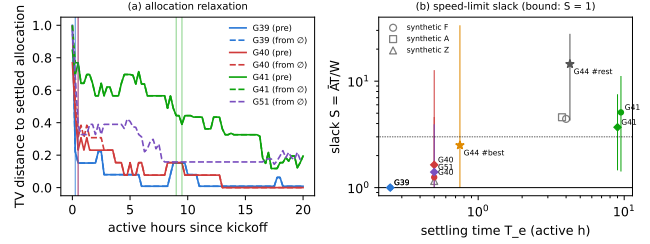
**Bottom line.** A kickoff that names its target re-allocates the swarm at the Wasserstein speed limit within half an active hour; a free-choice kickoff settles 5–40 times slower through churn, at the same switch rate.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



## WHAT HAPPENED

After a goal change, agents move their work to new repositories. I measured this move on DQ4 work commits in four regime-III kickoffs (G39, G40, G41, G51) and tested it against the Wasserstein speed limit  $T \geq W/\bar{A}$ . The synthetic check came first. In three kickoffs that name a target, the allocation freezes within 15–30 active min. Each moved agent switches once ( $S = \bar{A}T/W = 1.00\text{--}1.40$ ), so the move sits on the bound. G41 (no shared target) takes 9.5h with  $S = 5.1$  [1.4, 11.2]. In G44 the room with a named target settles  $\times 5.7$  faster than the free-choice room at the same switch rate. The field-limited predictions (slack  $\geq 3$ , settling 2.5–10 h) failed in 3/4 periods. Eleven newcomers settle in a median 0.28 h. The cadence elasticity is not identifiable at  $N \leq 25$ . Holdout (NE20): not run.



**Observables.** (a) Distance of the repo allocation to its settled state after the kickoff. G39, G40 and G51 drop within 0.5 active h; G41 relaxes over 10 h. (b) Slack  $S = \bar{A}T/W$  against the settling time  $T_e$ , with agent-bootstrap CIs.  $S = 1$  is the speed limit. Grey: synthetic field-limited (F), activity-limited (A) and instant-freeze (Z) means. Stars: the two G44 rooms on the same days.

## SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Useful as a design rule for kickoffs, with a small evidence base. If the kickoff names the artifact, the swarm re-points in 15–30 active min with no wasted switches. If agents must choose, re-pointing takes 4–10 h and 5–15 switches per net move. An operator who wants a fast, clean switch should name the repository or deliverable in the goal text. The speed-limit language adds little beyond this: the bound holds by counting, and the slack measures how specific the field is, not a thermodynamic cost. Whether faster call cadence speeds re-allocation is untested here; NE20 in the holdout can answer it.

## PHYSICAL MODEL

Models 02 (kinetic jump process) and 05 (driven population). Each agent of a period is one copy of a Markov jump process on repos plus a null state;  $p(t)$  is the share of agents on each repo. The kickoff is the field.  $W$ : total-variation distance between the allocations at  $t = 0^-$  and at the settling time  $T$ .  $\bar{A}$ : switches per agent per active hour.  $S$ : slack.  $\Sigma_{\text{ex}}$ : excess entropy production (Kolchinsky et al. 2026).

$$\dot{p} = \nabla^\top j, \quad A = \sum_\rho j_\rho, \quad W = \frac{1}{2} \|p(T) - p(0)\|_1$$

$$\Sigma_{\text{ex}}(T) \geq 2W \tanh^{-1}(W/\bar{A}T) \Rightarrow T \geq \frac{W}{\bar{A}} \coth \frac{\Sigma_{\text{ex}}}{2W} \geq \frac{W}{\bar{A}}$$

$$S = \bar{A}T/W \quad (\geq 1; S = 1 : \text{every switch directed})$$

## RESULTS VS PREDICTIONS

| Prediction                           | Observed                                                 | Verdict       |
|--------------------------------------|----------------------------------------------------------|---------------|
| P0 $\bar{A}T \geq W$                 | no violation (counting identity)                         | holds         |
| P1 $S \geq 3$ in $\geq 2/3$          | 1.00, 1.25, 5.09, 1.40                                   | fail (1/4)    |
| P2 $R \geq 3$ in $\geq 2/3$          | 0.01, 0.00, 6.9, 0.31                                    | fail (1/4)    |
| P3 $T_e \in [2.5, 10]$ h             | 0.25, 0.5, 9.5, 0.5 h                                    | fail; R2 wins |
| P4 pooled $\varepsilon$ vs synthetic | -0.12 [-2.8, 2.6]                                        | inconclusive  |
| N1 G44 rooms                         | $T$ ratio 5.7 vs activity rival 0.98                     | supported     |
| N2 newcomers, NE38                   | median 0.28 h; $\varepsilon = +1.3 \pm 1.3$ ; NE38 2.0 h | mixed         |
| tail slack $S_{90}$                  | 1.15, 5.9, 8.9, 4.0                                      | descriptive   |

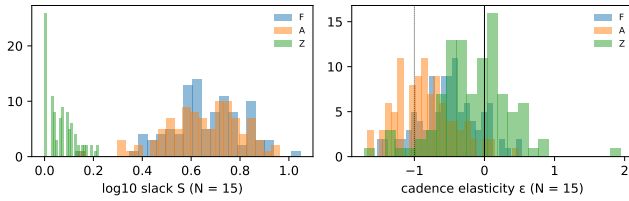
## GOAL PERIODS IN DETAIL

G39 × failed  
G40 × failed  
G41 ~ mixed  
G44 ✓ supported  
G51 × failed  
NE38 ~ mixed

## FAITHFULNESS SCORECARD

A mapping 1 F identifiability 1  
B assumptions 1 G ground truth 1  
C adequacy 1 H comparative 1  
D unfitted predictions 1 I transfer 1  
E interventional 1

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writup/paper.tex`).



**Second figure.** Synthetic agents observed only at their commits ( $N = 15$ , 100 replicates per scenario). Left: the slack separates an instant freeze (Z,  $S \approx 1.15$ ) from churn-dominated settling. It does not separate field-limited (F) from activity-limited (A) settling, because both give  $S \approx 4.5$ . Right: the cadence elasticity is biased ( $-0.5$  under F) and overlaps A.

## CAVEATS

The bound holds by construction on the empirical ensemble, so P0 is a check, not a test. Commits are the only observation of the state: the settling time includes the first-commit delay, and the 15-min grid floors  $T_e$  at 0.25 h. G51 has no pre-kickoff allocation (the previous week is held out), so its  $W$  counts every first commit as a move. Each period gives one kickoff with 13–16 agents, and bootstrap CIs of  $S$  are wide outside G39. G44’s named-target room has 4 committing agents. The cadence elasticity is not identified.

## ROUND 2

Score every regime-III kickoff for a named artifact and test  $S$  and  $T_e$  against it. Use the denser `artifact.mentions` channel to resolve sub-15-min settling. Run `confirm.py` (NE20 cadence DiD and transfer to #45–#47).

- **Essence.** an operator rule for how fast a swarm can be re-pointed, and whether the limit is the field, the agents’ switching, or their cadence.
- **H75-R1** Kickoff text vs slack across all regime-III kickoffs.
- **H75-R2** A denser allocation channel.
- **H75-R3** Excess EP of re-allocation.
- **H75-R4** Run `confirm.py` (NE20 DiD and transfer).

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